

Alpha capture curves provide managers with the following (Table 11.6):

1. Expected alpha capture (profit level) for a specified order size and alpha forecast. For example, if the manager wants to transact an order for 30% ADV in a stock where the forecasted alpha is 3% over three days, the maximum profit (excluding trading costs) the manager can expect to achieve is 148 bp.
2. It provides the manager with the the maximum order size that can be traded for a profit. For example, if the alpha forecast is 1% over three days the manager can trade up to 25% ADV. At 25% ADV the manager will net a profit of 2 bp. Trading more shares will cause the manager to incur a loss on the trade (e.g., the trading cost was higher than the alpha forecast). If the manager sets out to trade 30% ADV in this stock they would expect to incur a cost of -10 bp.
3. It provides the manager with the means to determine the appropriate order size while evaluating the economic opportunity cost of the trade. For example, following the scenario in Example 6, a manager whose most attractive investment opportunity is a stock with an alpha forecast expectation of 3% over three days and second most attractive opportunity is a stock with an alpha forecast of 2% over three days could trade up to 15% ADV in the first stock before having to allocate dollars to the second stock. This is shown in Table 11.6. Notice that for an order of 15% ADV, the maximum trading profit for the investment vehicle at 3% over three days is 201 bp. Investing any more than 15% ADV will cause the net profit level to be less than 200 bp (2%). Hence, the manager would be better off investing the incremental dollars in the next most attractive investment.

Important Note: Alpha capture curves provide managers with a quick reference for profitability. In our examples above, we only included the one-way implementation cost of the trade. The expectation in this analysis is that the manager would hold the acquired position over a longer period of time. Investors who are looking to take advantage of a short-term trend and trade in and out of these positions in shorter time horizons will also need to include the liquidation cost (sell cost) of the trade. If the expectation is that the market conditions will be exactly the same during liquidation of the order investors can simply double the implementation cost (buy cost). But if the market conditions are expected to be different during liquidation than acquisition managers could use techniques provided above to determine realistic liquidation costs for the order based on their expectation for market conditions during these times. Yet another reason why managers need their own market impact models. These expectations can be incorporated into the optimization process described above.